



# Hao Zhang

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Dr. Hao Zhang is currently a Postdoctoral Fellow at the Digital Future Institute, Khalifa University, working with Prof. Mérouane Debbah. He received his Ph.D. degree in Information and Communication Engineering from Nanjing University of Aeronautics and Astronautics (NUAA) in June 2025 and was selected for NUAA's Top-notch Postgraduate Innovative Talents Training "Yinhang Project."

His research focuses on **foundation models, signal processing, and wireless networks**. He has authored over **30 journal and conference papers** in leading venues such as *IEEE COMST*, *IEEE JSAC*, *IEEE TIFS*, *IEEE TCOM*, and *IEEE TCCN*. His work has received **900+ Google Scholar citations** with an **h-index of 18**, including **two ESI Highly Cited Papers** and an **cover story**. He received the **2025 Excellent Doctoral Dissertation Award of China Education Society of Electronics (CESE)**.

## Academic Appointments

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|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Present   | <b>Postdoctoral Fellow</b> , Digital Future Institute, Khalifa University                                                                      |
| Jun. 2026 | ► Researching foundation models for spectrum intelligence with Prof. Mérouane Debbah.                                                          |
| Jun. 2026 | <b>Assistant Researcher &amp; Postdoctoral Fellow</b> , College of Artificial Intelligence, Nanjing University of Aeronautics and Astronautics |
| Jun. 2025 | ► Research on spectrum foundation models and intelligent wireless signal processing.                                                           |

## Education

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|-----------|----------------------------------------------------------------------------------------------------------------------------|
| Jun. 2025 | College of Electronic & Information Engineering, <b>Nanjing University of Aeronautics and Astronautics</b>                 |
| Apr. 2021 | Ph.D. (Supervisor: Prof. Fuhui Zhou; Research Assistant, Jul. 2020–Mar. 2021) in Information and Communication Engineering |
| Dec. 2024 | School of Electrical & Electronic Engineering, <b>Nanyang Technological University</b>                                     |
| Jan. 2024 | Visiting Ph.D. Student (Supervisor: Prof. Chau Yuen, IEEE Fellow) in Information and Communication Engineering             |
| Jun. 2020 | School of Information Engineering, <b>Nanchang University</b>                                                              |
| Sep. 2017 | M.Eng. in Electrical and Communication Engineering                                                                         |
| Jun. 2017 | School of Information Engineering, <b>Nanchang University</b>                                                              |
| Sep. 2013 | B.Eng. in Internet of Things Engineering                                                                                   |

## Publications

### First-Author Journal Articles

- [1] **H. Zhang**, F. Zhou, H. Du, Q. Wu, and C. Yuen, "Revolution of wireless signal recognition for 6G: Recent advances, challenges and future directions," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 3521–3563, 2026, **ESI Highly Cited Paper**. DOI: 10.1109/COMST.2025.3569427.
- [2] **H. Zhang**, F. Zhou, Q. Wu, and C. Yuen, "Distributed multi-task learning for joint wireless signal enhancement and recognition," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 1554–1568, 2026. DOI: 10.1109/TCCN.2025.3612750.
- [3] **H. Zhang**, J.-J. Xu, H.-W. Cui, L. Li, Y. Yang, C.-S. Tang, and N. Boers, "When geoscience meets foundation models: Towards general geoscience artificial intelligence system," *IEEE Geoscience and Remote Sensing Magazine*, vol. 13, no. 4, pp. 79–118, 2025. DOI: 10.1109/MGRS.2024.3496478.

- [4] **H. Zhang**, F. Zhou, W. Wang, Q. Wu, and C. Yuen, "A federated learning-based lightweight network with zero trust for UAV authentication," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 7424–7437, 2025. DOI: 10.1109/TIFS.2025.3587624.
- [5] **H. Zhang**, F. Zhou, Q. Wu, and C. Yuen, "FSOS-AMC: Few-shot open-set learning for automatic modulation classification over multipath fading channels," *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 18 718–18 731, 2025. DOI: 10.1109/JIOT.2025.3557973.
- [6] **H. Zhang**, F. Zhou, Q. Wu, and C. Yuen, "Spectrum cognition: Semantic situation for next-generation spectrum management," *IEEE Network*, 2025. DOI: 10.1109/MNET.2025.3604901.
- [7] **H. Zhang**, F. Zhou, Q. Wu, and N. Al-Dhahir, "SSwsrNet: A semi-supervised few-shot learning framework for wireless signal recognition," *IEEE Transactions on Communications*, vol. 72, no. 9, pp. 5823–5836, 2024. DOI: 10.1109/TCOMM.2024.3385921.
- [8] **H. Zhang**, F. Zhou, Q. Wu, W. Wu, and R. Q. Hu, "A novel automatic modulation classification scheme based on multi-scale networks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 8, no. 1, pp. 97–110, 2022. DOI: 10.1109/TCCN.2021.3091730.
- [9] **H. Zhang**, X. Hong, and L. Zhu, "Detecting small objects in thermal images using single-shot detector," *Automatic Control and Computer Sciences*, vol. 55, no. 2, pp. 202–211, 2021. DOI: 10.3103/S0146411621020097.
- [10] **H. Zhang**, L. Yuan, G. Wu, F. Zhou, and Q. Wu, "Efficient automatic modulation classification using involution based residual networks," *IEEE Wireless Communications Letters*, vol. 10, no. 11, pp. 2417–2420, 2021. DOI: 10.1109/LWC.2021.3102069.
- [11] **H. Zhang**, and X. Hong, "Recent progresses on object detection: A brief review," *Multimedia Tools and Applications*, vol. 78, pp. 27 809–27 847, 2019, **CCF-C**. DOI: 10.1007/s11042-019-07898-2.

#### Co-first- and Co-authored Journal Articles

- [12] J.-J. Xu, **H. Zhang**, C.-S. Tang, Y. Yang, M. R. El-Maarry, L. Li, and B. Shi, "Drying induces mars intermediate-sized cracks: New evidence and insight from geometrical quantification," *Icarus*, p. 117 026, 2026, **Cover Story**, *Co-first author*. DOI: 10.1016/j.icarus.2026.117026.
- [13] J.-J. Xu, **H. Zhang**, C.-S. Tang, L. Li, and B. Shi, "XGeoS-AI: An interpretable learning framework for deciphering geoscience image segmentation," *Environmental Earth Sciences*, vol. 84, no. 3, p. 97, 2025. DOI: 10.1007/s12665-025-12095-6.
- [14] S. Zhao, Q. Wu, F. Zhou, **H. Zhang**, Y. Huang, and K.-K. Ma, "Cognitive escape reinforcement learning for complex decision making," *IEEE Transactions on Vehicular Technology*, pp. 1–13, 2025. DOI: 10.1109/TVT.2025.3597537.
- [15] F. Zhou, C. Liu, **H. Zhang**, W. Wu, Q. Wu, T. Q. S. Quek, and C.-B. Chae, "SpectrumFM: A foundation model for intelligent spectrum management," *IEEE Journal on Selected Areas in Communications*, 2025. DOI: 10.1109/JSAC.2025.3644783.
- [16] J.-J. Xu, C.-S. Tang, Y. Yang, L. Li, **H. Zhang**, Q. Cheng, X.-Y. Zhang, B. Liu, and B. Shi, "Breathing phenomenon of soil desiccation cracking: Insights from novel geophysical observations," *Journal of Geophysical Research: Earth Surface*, vol. 129, no. 1, e2023JF007318, 2024. DOI: 10.1029/2023JF007318.
- [17] J.-J. Xu, **H. Zhang**, C.-S. Tang, Y. Yang, L. Li, D.-L. Wang, B. Liu, and B. Shi, "Soil desiccation crack recognition: New paradigm and field application," *Journal of Geophysical Research: Machine Learning and Computation*, vol. 1, no. 3, e2024JH000347, 2024, *Co-first author*. DOI: 10.1029/2024JH000347.
- [18] J.-J. Xu, **H. Zhang**, C.-S. Tang, Q. Cheng, B. Liu, and B. Shi, "Automatic soil desiccation crack recognition using deep learning," *Géotechnique*, vol. 72, no. 4, pp. 337–349, 2022, **ESI Highly Cited Paper** & 75th Géotechnique Anniversary Early Career Award. DOI: 10.1680/jgeot.20.P.091.
- [19] J.-J. Xu, **H. Zhang**, C.-S. Tang, Q. Cheng, B.-g. Tian, B. Liu, and B. Shi, "Automatic soil crack recognition under uneven illumination condition with the application of artificial intelligence," *Engineering Geology*, vol. 296, p. 106 495, 2022. DOI: 10.1016/j.enggeo.2021.106495.
- [20] L. Yuan, **H. Zhang**, M. Xu, F. Zhou, and Q. Wu, "A multi-scale CNN framework for wireless technique classification in beyond 5G communications," *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 10 366–10 367, 2022. DOI: 10.1109/JIOT.2021.3132652.
- [21] Q. Zhou, **H. Zhang**, and S. Wang, "Artificial intelligence, big data, and blockchain in food safety," *International Journal of Food Engineering*, vol. 18, no. 1, pp. 1–14, 2022. DOI: 10.1515/ijfe-2021-0299.

## First-Author Conference Papers

- [22] **H. Zhang**, F. Zhou, Q. Wu, and C. Yuen, "FSOS-AMC: Few-shot open-set learning for automatic modulation classification," *Proc. 16th Int. Conf. Wireless Communications and Signal Processing (WCSP)*, 2024, pp. 746–751. DOI: 10.1109/WCSP62071.2024.10826902.
- [23] **H. Zhang**, X. Hong, S. Zhou, and Q. Wang, "Infrared image segmentation for photovoltaic panels based on Res-UNet," *Proc. Chinese Conf. Pattern Recognition and Computer Vision (PRCV). LNCS*, vol. 11857, **CCF-C**, 2019, pp. 611–622. DOI: 10.1007/978-3-030-31654-9\_52.

## Co-authored Conference Papers

- [24] X. Huang, X. Huang, C. Liu, **H. Zhang**, F. Zhou, and Q. Wu, "From survey to design: Knowledge-enhanced multimodal spectrum foundation model for intelligent spectrum management," *Proc. 3rd Int. Conf. Ubiquitous Communication (Ucom)*, 2025, pp. 231–236. DOI: 10.1109/Ucom67224.2025.11336912.
- [25] C. Liu, **H. Zhang**, F. Zhou, W. Wu, Q. Wu, D. W. K. Ng, T. Q. S. Quek, and C.-B. Chae, "SpectrumFM: Redefining spectrum cognition via foundation modeling," *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2025, pp. 3861–3866. DOI: 10.1109/GLOBECOM59602.2025.11432674.
- [26] Z. Tong, **H. Zhang**, M. Xu, and F. Zhou, "An accurate few-shot spectrum anomaly detection method using the contrastive language-image pre-training model," *Proc. 3rd Int. Conf. Ubiquitous Communication (Ucom)*, 2025, pp. 220–224. DOI: 10.1109/Ucom67224.2025.11336960.
- [27] Y. Wei, X. Liu, **H. Zhang**, F. Zhou, and Q. Wu, "GenSpectralLM: Large model-driven spectrum map construction with electromagnetic propagation," *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2025, pp. 393–398. DOI: 10.1109/GLOBECOM59602.2025.11431695.
- [28] M. Xu, H. Ma, **H. Zhang**, F. Zhou, and Q. Wu, "PLMSNet: A pseudo-labeling multi-scale network for semi-supervised spectrum sensing," *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2025, pp. 3915–3920. DOI: 10.1109/GLOBECOM59602.2025.11432737.
- [29] D. Zhang, X. Yan, H. Ma, M. Xu, **H. Zhang**, and F. Zhou, "YOLOv10-P2: A robust detection scheme for UAV signals in RF spectrograms," *Proc. 3rd Int. Conf. Ubiquitous Communication (Ucom)*, 2025, pp. 249–253. DOI: 10.1109/Ucom67224.2025.11336824.
- [30] D. Han, **H. Zhang**, S. Wang, W. Chai, H. Zhou, and F. Zhou, "Small objects recognition by exploiting an improved YOLOv5 algorithm on the UAV platform," *Proc. Int. Conf. Ubiquitous Communication (Ucom)*, 2023, pp. 193–198. DOI: 10.1109/Ucom59132.2023.10257606.
- [31] R. Wang, **H. Zhang**, M. Xu, F. Zhou, and Q. Wu, "A novel lightweight automatic modulation classification scheme based on inverted residuals," *Proc. Int. Conf. Ubiquitous Communication (Ucom)*, 2023, pp. 259–263. DOI: 10.1109/Ucom59132.2023.10257638.
- [32] R. Ding, **H. Zhang**, F. Zhou, Q. Wu, and Z. Han, "Data-and-knowledge dual-driven automatic modulation recognition for wireless communication networks," *Proc. IEEE International Conference on Communications (ICC)*, 2022, pp. 1962–1967. DOI: 10.1109/ICC45855.2022.9838977.
- [33] L. Hu, Y. Li, **H. Zhang**, L. Yuan, F. Zhou, and Q. Wu, "Robust semantic communication driven by knowledge graph," *Proc. 9th Int. Conf. Internet of Things: Systems, Management and Security (IOTSMS)*, 2022, pp. 1–5. DOI: 10.1109/IOTSMS58070.2022.10061867.
- [34] M. Xu, Y. Wu, **H. Zhang**, L. Yuan, Y. Wan, F. Zhou, and Q. Wu, "GAN-enabled robust backdoor attack for UAV recognition," *Proc. 7th Int. Conf. Communication, Image and Signal Processing (CCISP)*, 2022, pp. 474–478. DOI: 10.1109/CCISP55629.2022.9974216.
- [35] Q. Wang, **H. Zhang**, X. Hong, and Q. Zhou, "Small object detection based on modified FSSD and model compression," *Proc. 6th IEEE Int. Conf. Signal and Image Processing (ICSIP)*, 2021, pp. 88–92. DOI: 10.1109/ICSIP52628.2021.9688896.

## Preprints

- [36] Y. Wei, X. Liu, **H. Zhang**, Y. Diao, H. Wu, L. Cao, F. Zhou, and Q. Wu, "Genspectralm-3d: A physics-informed foundation model for 3d electromagnetic spectrum map construction," *IEEE Internet of Things Journal*, Under Review, 2026.

[37] M. Xu, X. Yan, **H. Zhang**, F. Zhou, and Q. Wu, “Knowledge enhanced intelligent wireless signal detection and classification for coexisting and overlapping signals,” *IEEE Transactions on Cognitive Communications and Networking*, Under Review, 2026.

 **Patents & Standards**

1. **Hao Zhang**, Fuhui Zhou, Jiaxin Ding, Liang Chang, Shengmei Luo, Zhihong Lu, and Qihui Wu, “A semi-supervised intelligent and accurate recognition method for small sample wireless signals,” Chinese Patent Application Publication No. CN119128666A.
2. Yifan Wei, Xiaodong Liu, **Hao Zhang**, Fuhui Zhou, Qihui Wu, Anping Li, and Liang Chang, “A method for constructing spectral situation based on a large electromagnetic model,” Chinese Patent Application Publication No. CN121547133A.
3. Fuhui Zhou, Rui Ding, Ming Xu, **Hao Zhang**, Lu Yuan, Qihui Wu and Chao Dong, “Intelligent data and knowledge-driven method for modulation recognition,” U.S. Patent No. 11,700,156, Jul. 2023.
4. Fuhui Zhou, Dongming Li, Junchang Chen, **Hao Zhang**, and Wei Wu, “Proposed Revision of the Handbook on Spectrum Monitoring (PROPOSED REVISIONS TO SECTION 4.10 OF THE ITU SPECTRUM MONITORING HANDBOOK),” ITU-R R23-WP1C, 1C-RG-SMH/35-E, Oct. 2024.

 **Projects**

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|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun. 2028 | Research on Physics-Driven Deep Understanding Methods of Electromagnetic Spectrum                                                                                                                                                                      |
| Jun. 2026 | ▶ General Program of China Postdoctoral Science Foundation (Grant 2026M794729; CNY 80,000), <b>Project Lead</b> .                                                                                                                                      |
| Dec. 2024 | Few-Shot Modulation Identification in Highly Dynamic Environments                                                                                                                                                                                      |
| Jun. 2023 | ▶ Led the design and evaluation of few-shot and open-set modulation recognition methods for rapidly varying wireless channels.<br>▶ Postgraduate Research & Practice Innovation Program of Jiangsu Province (Grant KYCX23_0380), <b>Project Lead</b> . |
| Jun. 2024 | Mechanical Performance of Coral Concrete Wind Turbines Based on Interpretable Deep Learning                                                                                                                                                            |
| Jun. 2023 | ▶ Co-led an interdisciplinary study applying interpretable deep learning to assess coral-concrete wind-turbine structures.<br>▶ Interdisciplinary Innovation Fund for Doctoral Students of NUAU (Grant KXKCXJJ202302), <b>Project Co-Lead</b> .        |

 **Honors**

- ▶ **Excellent Doctoral Dissertation Award of China Education Society of Electronics (CESE)**, Dec. 2025
- ▶ Outstanding Graduate, Nanjing University of Aeronautics and Astronautics, Dec. 2024
- ▶ Chinese Government Scholarship, China Scholarship Council (CSC), Jul. 2023
- ▶ Top-notch Postgraduate Innovative Talents Training “Yinhang Project” of NUAU, May 2022
- ▶ Outstanding Graduate of Nanchang University (top 4%), Jun. 2020
- ▶ Graduate Scholarships, Nanchang University: First Class (2019, 2020) and Second Class (2018)
- ▶ Undergraduate Scholarships, Nanchang University: Special Class (2015), First Class (2014, 2016, 2017), and Second Class (2014)

 **Other**

- ▶ **Memberships & Service:** IEEE Member; ICC 2025 SAC TPC Member; VTC2024-Spring Session Chair
- ▶ **Peer Review:** Reviewer for 18 journals (e.g., IEEE JSAC, TNNLS, TIFS, TWC, TCOM, and TCCN) and major conferences (ACM MM, IEEE GLOBECOM, ICC, VTC, and WCSP)